

Nursing fundamentals

A visual knowledge pack

Core concepts. Bedside observations.
Medication safety. Clear escalation cues.



LET'S WORK THROUGH IT

Assess.
Understand.
Give safely.

Look for my notes and pause to connect each concept with your nursing assessment.

45 illustrated topic sheets

How to use this pack

A revision companion for nursing students and practising nurses.



READ ONE SHEET AT A TIME

Start with the core concept. Link it to what you would assess, what a medicine should achieve and what would make you seek help.

USE IT WITH LOCAL GUIDANCE

This is an educational resource, not a prescribing or procedural protocol. Patient-specific orders, your scope of practice and current local policies take precedence.

AUSTRALIAN LANGUAGE

Glucose is in mmol/L; haemoglobin in g/L. Medicine names include paracetamol, salbutamol and glyceryl trinitrate. Older or overseas synonyms are explained where helpful.

FOLLOW THE SOURCES

Source codes at the foot of each sheet are clickable. The reference directory gives full titles and links. Clinical guidance and product availability can change.

VISUAL KEY

Follow the cartoon nurse for a key takeaway on each topic. Pause-and-think prompts connect the facts to your assessment. Red ACT / CHECK notes highlight risks and escalation.

Evidence basis: Australian standards, guidelines and medicines information, with textbook anatomy and physiology. This synthesis has not had independent clinical peer review; review it before adopting it as a workplace teaching resource.

Find a topic

Assessment, care and core concepts | Sheet numbers and PDF pages



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Diabetes, respiratory care and medicines | Sheet numbers and PDF pages



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Assess the whole person

A structured first look, followed by a focused assessment.



START WITH ABCDE

- A Airway: patency, voice, obstruction.
- B Breathing: rate, effort, SpO₂, oxygen device.
- C Circulation: pulse, BP, perfusion, bleeding.
- D Disability: consciousness, pupils, glucose.
- E Exposure: temperature, skin, wounds; preserve dignity.

HEAD TO TOE

- Pain and symptoms; cognition and communication; heart and lungs; abdomen, nutrition and swallowing; urine and bowels; skin and pressure areas; movement, falls risk and devices.

MAKE IT PERSON-CENTRED

- Confirm identity and consent.
- Ask what matters to the person.
- Use an interpreter when needed.
- Compare findings with their usual function, goals of care and documented baseline.

CLOSE THE LOOP

- Record findings and trends.
- Act on abnormalities.
- Explain the plan, check understanding and reassess after interventions.
- Confirm call bell, mobility aids and immediate safety.



NURSE'S NOTE

A change from baseline can matter before a number crosses a threshold.

ACT / CHECK

Call for urgent help for airway compromise, severe breathing difficulty, shock or a new major neurological change.

Observations & documentation

Measure accurately. Describe what you found. Record what happened next.



CORE OBSERVATIONS

- Respiratory rate, SpO₂ and oxygen therapy.
- Heart rate and blood pressure.
- Temperature and consciousness.
- Pain.
- Glucose, fluid balance and other measures when indicated.

INTERPRET IN CONTEXT

- Resting adult pulse: 60–100/min.
- Resting respiratory rate: 12–20/min.
- These are learning guides, not escalation limits.
- Use the local observation chart, prescribed targets and trends.

DOCUMENT OBJECTIVELY

- Date and time.
- Relevant symptoms and actual measurements.
- Assessment, action and response.
- Who was contacted, when, advice received and follow-up.
- Use approved abbreviations.

HAND OVER WITH ISBAR

- I – Identify yourself and the patient.
- S – State the immediate situation.
- B – Give relevant background.
- A – Share assessment and concern.
- R – Request a clear action and timeframe.



NURSE'S NOTE

Example: "14:10 RR 28/min, increased effort; RN in charge notified; urgent review requested."

ACT / CHECK

Escalate clinical concern even when the observation score is low. Confirm the documented resuscitation plan.

Consciousness & GCS

Check for barriers to testing, then record each component.



Component	Response and score
EYE OPENING / E	4 Spontaneous 3 To sound 2 To pressure 1 None
VERBAL RESPONSE / V	5 Orientated 4 Confused 3 Words 2 Sounds 1 None
MOTOR RESPONSE / M	6 Obeys commands 5 Localising 4 Normal flexion 3 Abnormal flexion 2 Extension 1 None

PAUSE & THINK What would you assess first, and what change would you escalate?



NURSE'S NOTE

Report E, V and M individually, e.g. E4 V5 M6 = 15/15. Total range is 3–15 when all parts are testable.

ACT / CHECK

A falling score or new confusion needs prompt assessment. Record "not testable" and the reason when appropriate; do not invent a score.

PPE & hand hygiene

Choose protection for the task and the transmission risk.



PUTTING ON

1. Hand hygiene.
2. Gown.
3. Surgical mask or P2/N95 respirator as indicated.
4. Protective eyewear or face shield.
5. Gloves over gown cuffs.

TAKING OFF

1. Remove gloves.
2. Hand hygiene.
3. Remove gown; hand hygiene.
4. Remove eye protection by straps/arms; hand hygiene if contaminated.
5. Remove mask by ties/loops; hand hygiene.

RESPIRATOR CHECK

- Use the correct fit-tested respirator.
- Perform a fit check each time.
- Airborne precautions affect where respirators are removed; follow the isolation-room procedure.

CLEAN HANDS STILL MATTER

- Gloves do not replace hand hygiene.
- Avoid touching your face, phone or clean surfaces with used gloves.
- Remove PPE without touching contaminated outer surfaces.



NURSE'S NOTE

Gloves and gown may be removed together under the approved local sequence.

ACT / CHECK

Follow the displayed isolation instructions. Special pathogens and procedures may require a different PPE sequence.

Wounds & exudate

Describe drainage alongside the wound, surrounding skin and patient.



Appearance	What to record
Serous	Clear or pale yellow; thin and watery.
Haemoserous	Pink, thin drainage: serum mixed with a small amount of blood.
Sanguineous	Predominantly blood. Assess whether amount and timing are expected.
Purulent	Thick, opaque drainage; may be yellow, green or brown. Assess for infection.

ASSESS THE WHOLE WOUND

- Location, dimensions, wound bed, edges, drainage amount/odour, pain and surrounding skin.
- Compare with previous assessments and document the dressing plan.

LOOK BEYOND COLOUR

- Increasing pain, warmth, swelling, spreading erythema, malodour, wound breakdown or systemic illness warrant review.
- Protect skin from excess moisture and address pressure or perfusion problems.



NURSE'S NOTE

The amount and trend of exudate are as important as its colour.

ACT / CHECK

Heavy or persistent fresh bleeding, spreading infection or systemic deterioration requires urgent escalation.

Wounds: clinical photographs

Compare the wound bed, edges and surrounding skin.



ARTERIAL ULCER EXAMPLE

Visible features: an open wound on the foot, with blood visible on the gauze. Source identifies peripheral vascular disease.

Jonathan Moore / CC BY 3.0 / A9



VENOUS ULCER EXAMPLE

Visible features: open areas on the lower leg with surrounding colour and skin changes. Source identifies varicose ulcers.

Nini00 / CC BY-SA 3.0 / A10

Original photographs reproduced at reduced size; no clinical features altered. Image licences remain as credited. Full source links are in the reference directory.



NURSE'S NOTE

Describe what you see, then assess the patient. Photographs alone cannot establish infection, perfusion or the type and amount of drainage.

ACT / CHECK

Exudate must be assessed directly and over time. Do not label yellow wound-bed tissue as pus from a photograph alone.

Injection routes

The route, access device and medicine determine the technique.



Route	Core principle
Intravenous	Access devices: peripheral IV cannula; PICC; central venous catheter; implanted port. A midline is peripheral access. Check device suitability, patency and local policy.
Subcutaneous	Into subcutaneous tissue. Vaccine guidance uses 45°; insulin and other devices may use 90°.
Intramuscular	Into muscle; 90°. Correct landmarking and adequate needle length are essential.

BEFORE THE INJECTION

- Check the prescription, route, formulation, identity, allergy history, site and equipment.
- Explain the procedure.
- Choose needle length/gauge for the patient and product.

AFTER THE INJECTION

- Dispose of the needle immediately into a sharps container.
- Do not recap.
- Record administration and assess the response.
- Follow product-specific observation requirements.



NURSE'S NOTE

These angles are orientation guides, not a substitute for demonstrated injection competency.

ACT / CHECK

Never choose a site or needle from age alone. Verify landmarks and product instructions.

Peripheral IV care

Choose a suitable catheter and keep checking whether it is still needed.



Common colour coding*	Selection considerations
14 G / orange; 16 G / grey	Larger bore: rapid fluid or blood delivery when clinically required.
18 G / green; 20 G / pink	Common adult options; choose for the therapy, vein and flow required.
22 G / blue; 24 G / yellow	Smaller bore: may suit small or fragile veins when adequate for therapy.

BEFORE EACH USE

- Inspect and ask about pain.
- Check redness, swelling, leakage, dressing integrity and patency under local policy.
- Use aseptic technique and disinfect access points.

REMOVE WHEN UNNEEDED

- Review the indication regularly.
- Document site assessments and complications.
- Do not force a resistant flush.
- Catheter length, vein and fluid viscosity also affect flow.



NURSE'S NOTE

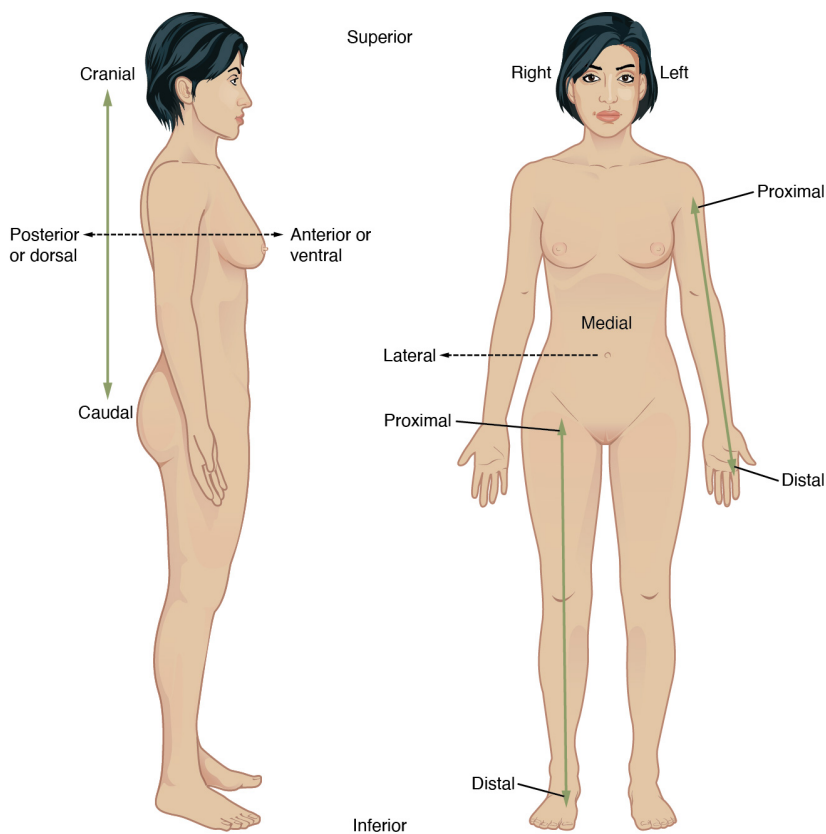
Lower gauge number = wider catheter. *Confirm the device label; colour alone is not identification.

ACT / CHECK

For suspected extravasation: stop the infusion, do not flush, and follow the drug-specific pathway before removing the catheter.

Anatomical directions

Describe position using the body in standard anatomical position.



Anterior / Posterior

Anterior: Toward the front.
Posterior: Toward the back.

Medial / Lateral

Medial: Toward the midline.
Lateral: Away from the midline.

Superior / Inferior

Superior: Higher up; toward the head.
Inferior: Lower down; toward the feet.

Proximal / Distal

Proximal: Closer to the point of attachment.
Distal: Farther from the point of attachment.

Standard anatomical position: upright, facing forward, arms at the sides, palms forward.

OpenStax, Anatomy and Physiology 2e, Figure 1.13. Original image and labels retained.

PAUSE & THINK Can you connect the structure to its function?



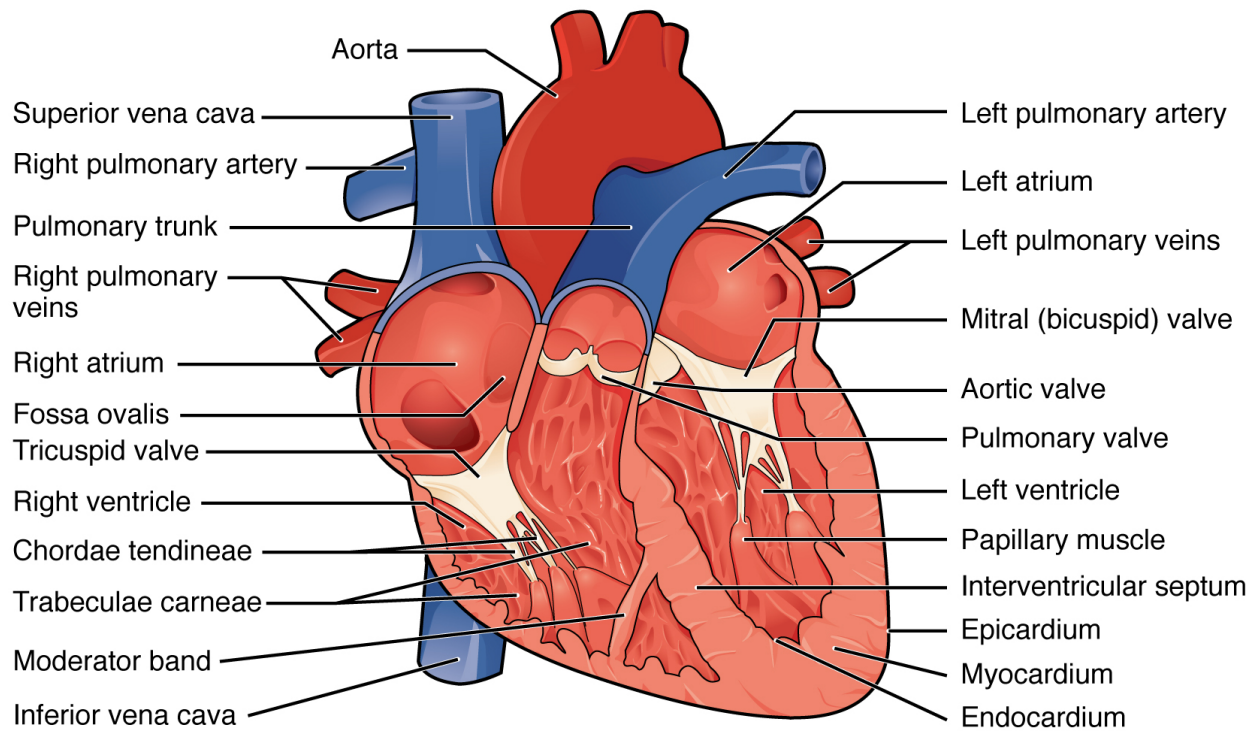
NURSE'S NOTE

Use the patient's anatomical right and left. Directional terms compare one structure with another.

Access for free at <https://openstax.org/books/anatomy-and-physiology-2e/pages/1-introduction>

Heart anatomy

Four chambers, four valves and one continuous circulation.



Anterior view

Textbook illustration: OpenStax, Anatomy and Physiology 2e. Original labels retained.

TRACE THE FLOW

- Body → venae cavae → right atrium → tricuspid valve → right ventricle → pulmonary valve → pulmonary arteries → lungs.

RETURN TO THE BODY

- Lungs → pulmonary veins → left atrium → mitral valve → left ventricle → aortic valve → aorta → body.



NURSE'S NOTE

Anatomical right and left refer to the patient. Pulmonary veins return blood to the **LEFT ATRIUM**.

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Blood & circulation

Arteries travel away from the heart; veins return towards it.



Component	Essential function
Red cells	Haemoglobin transports oxygen. Red cells are produced in bone marrow.
White cells	Immune defence. Different cell types have different roles.
Platelets	Cell fragments that contribute to the initial haemostatic plug.
Plasma	Water, proteins and dissolved substances. Most plasma proteins are made in the liver; antibodies are made by plasma cells.

PULMONARY CIRCUIT

- Right ventricle → lungs → left atrium.
- Pulmonary arteries usually carry deoxygenated blood and pulmonary veins oxygenated blood.

SYSTEMIC CIRCUIT

- Left ventricle → body → right atrium.
- Capillaries are exchange vessels; coronary arteries supply the heart muscle itself.



NURSE'S NOTE

Oxygenation does not define an artery or vein; direction of flow does.

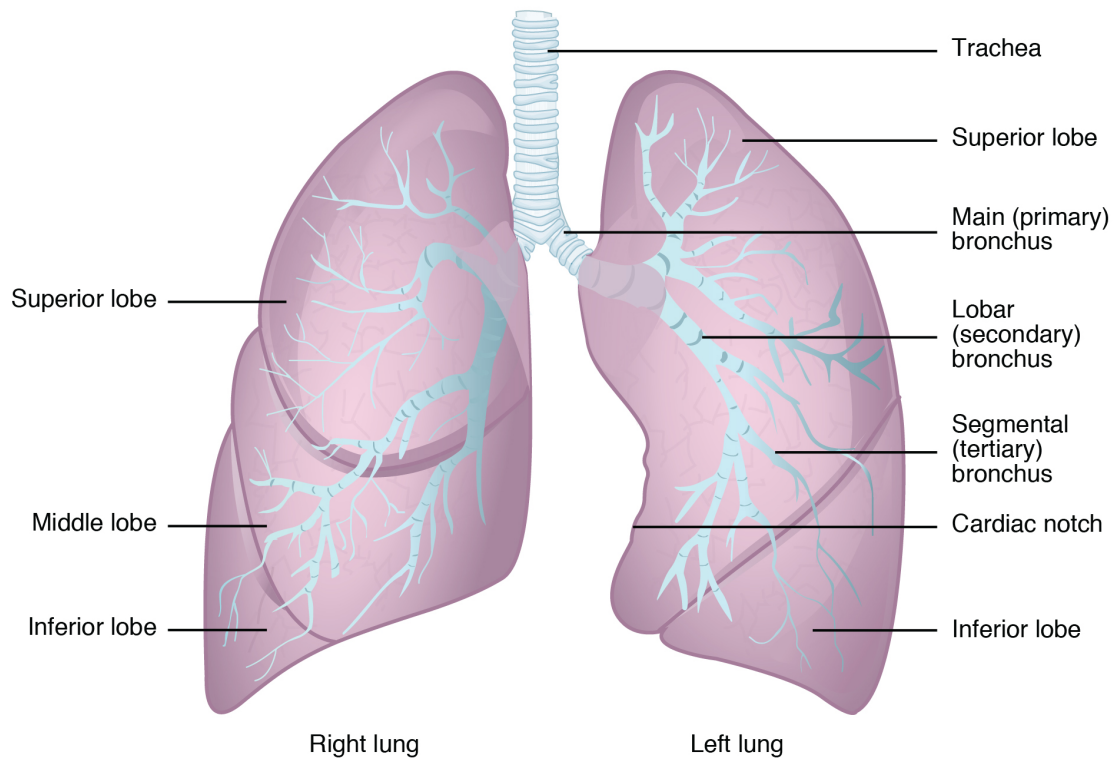
Access for free at <https://openstax.org/books/anatomy-and-physiology-2e/pages/1-introduction>

Sources: B1 B2

Full references at the back

Lung anatomy

Conducting airways move air; alveoli exchange gases.



Textbook illustration: OpenStax, Anatomy and Physiology 2e. Original labels retained.

AIRFLOW

- Trachea → main bronchi → smaller bronchi → bronchioles → alveoli.
- The right lung has three lobes; the left has two and a cardiac notch.

PLEURA & GAS EXCHANGE

- Visceral pleura covers the lung; parietal pleura lines the chest wall.
- Oxygen and carbon dioxide diffuse across the alveolar-capillary membrane.



NURSE'S NOTE

Ventilation is movement of air. Oxygenation is oxygen transfer into blood. One can fail despite the other.

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Breath sounds

Listen side-to-side and interpret sounds with effort, rate and SpO₂.



Sound	Clinical description
Vesicular	Soft, low-pitched breath sounds over most peripheral lung fields.
Fine crackles	Brief, fine pops, often on inspiration; can occur with interstitial fluid or fibrosis.
Coarse crackles	Louder bubbling/popping; may occur with airway secretions.
Wheeze	Musical, often expiratory; suggests narrowed airways.
Rhonchi	Low-pitched snoring/rattling; may change after coughing.
Pleural rub	Localised grating sound from inflamed pleural surfaces.
Stridor	Harsh, usually inspiratory sound suggesting upper airway narrowing.



NURSE'S NOTE

Document location, timing, symmetry and change after coughing or treatment. Sounds alone do not establish a diagnosis.

ACT / CHECK

New stridor, a silent chest with distress, or suddenly reduced air entry needs urgent assessment.

Oxygen therapy

Prescribe a target, choose the device, titrate and reassess.



Device	Key nursing point
Nasal cannulae	Standard low-flow guide: 1–4 L/min (upper end 4 L/min for this guide). Check the device and local policy; escalate if the target is not met.
Simple face mask	6–10 L/min. Follow device instructions; do not run below 5 L/min because of rebreathing risk.
Reservoir mask	Non-rebreather: typically 15 L/min initially in severe hypoxaemia. Keep reservoir inflated.
Venturi mask	Use the flow printed on the selected adaptor; delivers a specified oxygen concentration.
High-flow nasal oxygen	Heated, humidified system. Flow and FiO ₂ are set separately; requires appropriate monitoring.

TARGETS FOR ACUTE ADULT CARE

- Most adults: 92–96% (TSANZ).
- Risk of hypercapnic respiratory failure: usually 88–92%.
- Use the prescribed patient-specific range and blood-gas findings; 95% or above is not a universal oxygen-treatment target.



NURSE'S NOTE

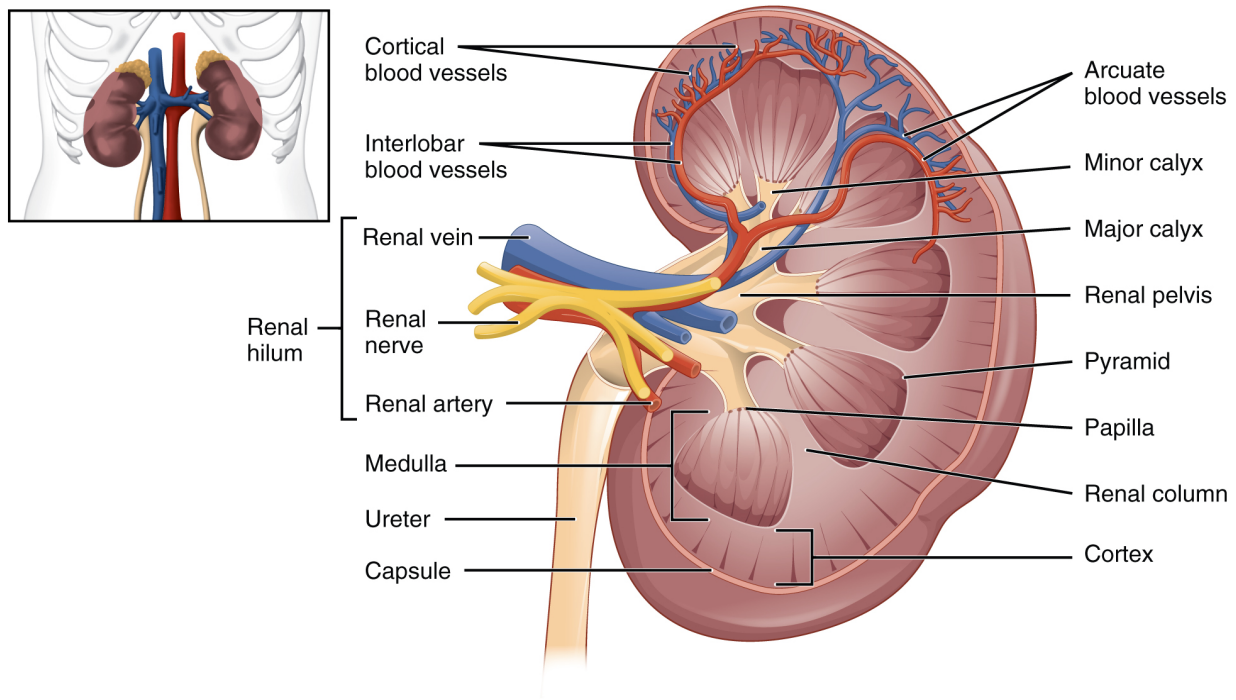
FiO₂ means inspired oxygen fraction. Low-flow device percentages are estimates, not guaranteed doses.

ACT / CHECK

Rising oxygen needs, drowsiness or worsening effort needs escalation. SpO₂ does not measure carbon dioxide.

Kidney anatomy

Filtration, fluid balance, electrolytes and medicine clearance.



Textbook illustration: OpenStax, Anatomy and Physiology 2e. Original labels retained.

STRUCTURE TO FUNCTION

- The cortex and medulla contain nephrons.
- Blood reaches the kidney through the renal artery and leaves through the renal vein.
- Urine drains via calyces → renal pelvis → ureter.

NURSING LINKS

- Trend urine output, creatinine/eGFR, electrolytes, weight and fluid balance.
- Reduced output may reflect poor perfusion, obstruction or kidney injury; assess the cause.



NURSE'S NOTE

A creatinine result must be interpreted with baseline, trend and clinical context. A "normal" value does not exclude low renal reserve.

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IV fluids & tonicity

First identify the problem: resuscitation, maintenance or replacement.



Fluid concept	What it means
Isotonic	0.9% sodium chloride or balanced crystalloids expand extracellular volume with little direct change in cell size.
Hypotonic	0.45% sodium chloride provides relatively more free water; can move water into cells. Not routine shock resuscitation.
Hypertonic	3% sodium chloride draws water out of cells. Used for specific indications such as symptomatic hyponatraemia or raised intracranial pressure.
Glucose 5%	Approximately isotonic in the bag, but behaves as free water after glucose is metabolised.

ASSESS AND REASSESS

- Check indication, prescribed fluid/rate, losses, oral intake, BP, perfusion, lung signs, oedema, weight, urine and electrolytes.
- Reassess after a bolus or any clinical change.



NURSE'S NOTE

Isotonic fluid mainly fills the extracellular compartment; it does not simply "refill the cells".

ACT / CHECK

Hypertonic saline is high risk. Rapid sodium correction can cause serious harm. Use specialist orders and monitoring.

Laboratory essentials

Adult learning ranges: the reporting laboratory remains the reference.



Test	Reference guide	Nursing link
Sodium	135–145 mmol/L	Water balance; assess symptoms and rate of change.
Potassium	About 3.5–5.2 mmol/L	Cardiac conduction; check haemolysis, renal function and medicines.
Platelets	150–400 × 10 ⁹ /L	Bleeding/clotting context; assess trends.
Haemoglobin	Use laboratory range (g/L)	Interpret against bleeding, symptoms, baseline and fluid status.
White cell count	Use laboratory range (× 10 ⁹ /L)	Review differential, infection signs and medicine effects.
Creatinine / eGFR	Use baseline and trend	eGFR is less reliable during rapidly changing kidney function.
Troponin	Assay-specific limit (often ng/L)	Interpret serial change with symptoms and ECG.



NURSE'S NOTE

A reference interval is not a treatment target or an automatic emergency threshold.

ACT / CHECK

Act promptly on critical results. A haemolysed high potassium result needs timely reassessment, not automatic dismissal.

ABG fundamentals

Read the pH, then the respiratory and metabolic components.



Measure	Typical adult guide	Interpretation
pH	7.35–7.45	Below: acidaemia. Above: alkalaemia.
PaCO ₂	35–45 mmHg	Higher CO ₂ has an acidifying effect.
HCO ₃ [–]	22–26 mmol/L	Lower bicarbonate has an acidifying effect.
PaO ₂	Context-dependent	Interpret with age, altitude and the oxygen device/FiO ₂ .

COMMON PATTERNS

Low pH + high PaCO₂:
Respiratory acidosis pattern

Low pH + low HCO₃[–]:
Metabolic acidosis pattern

High pH + low PaCO₂:
Respiratory alkalosis pattern

High pH + high HCO₃[–]:
Metabolic alkalosis pattern

CHECK THE WHOLE PICTURE

- Compensation and mixed disorders can alter these patterns.
- A normal pH can conceal a mixed disorder.
- VBG is useful for many acid-base questions but venous PO₂ cannot assess arterial oxygenation.



NURSE'S NOTE

Record oxygen delivery at sampling. Evaluate the person and trend, not just the blood-gas printout.

Cranial nerves

A compact function map for a focused neurological assessment.



Nerve	Main function
I / Olfactory	Smell
II / Optic	Vision; afferent limb of pupil light reflex
III / Oculomotor	Most eye movements, eyelid lift, pupil constriction
IV / Trochlear	Superior oblique muscle; downward gaze in adduction
V / Trigeminal	Facial sensation and chewing
VI / Abducens	Lateral eye movement
VII / Facial	Facial expression; taste anterior two-thirds of tongue
VIII / Vestibulocochlear	Hearing and balance
IX / Glossopharyngeal	Pharyngeal sensation; taste posterior third; swallowing
X / Vagus	Voice, palate, swallowing; parasympathetic functions
XI / Accessory	Shoulder shrug and head turn
XII / Hypoglossal	Tongue movement



NURSE'S NOTE

Assess according to indication and competency; document asymmetry and change from baseline.

ACT / CHECK

New facial droop, speech difficulty, weakness or visual change: activate the local stroke pathway.

Access for free at <https://openstax.org/books/anatomy-and-physiology-2e/pages/1-introduction>

Sources: B5

Full references at the back

Autonomic nervous system

Effects depend on the receptor and the target organ.



Target	Sympathetic	Parasympathetic
Heart	Usually increases rate and contractility	Slows rate and AV conduction
Pupils	Dilates	Constricts
Airways	Beta-2 activation promotes dilation	Promotes constriction and secretion
Gut	Generally reduces motility	Promotes motility and secretion
Bladder	Favours storage	Favours emptying

FIGHT OR FLIGHT

- Supports responses to activity and stress.
- Most sympathetic postganglionic fibres release noradrenaline; sweat glands are an important cholinergic exception.

REST AND DIGEST

- Supports digestion and conservation of energy.
- Acetylcholine acts at muscarinic receptors on many target organs.



NURSE'S NOTE

Most systemic blood vessels are mainly controlled by sympathetic tone; parasympathetic activity is not a universal vasodilator.

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Sources: B6 B7

Full references at the back

Cholinergic & anticholinergic

This comparison describes mainly muscarinic effects.



Target	Cholinergic effect	Antimuscarinic effect
Saliva	Increased secretion	Dry mouth
Bladder	Promotes emptying	Urinary retention
Bowel	Increased motility	Constipation
Heart	Slower rate	Tachycardia
Airways	Constriction; more secretions	Dilation; fewer secretions
Pupils	Constriction	Dilation; blurred vision

MEDICINE EXAMPLES

- Ipratropium/tiotropium act in the airways; oxybutynin in the bladder.
- Some antidepressants, antipsychotics and antihistamines also add anticholinergic effects.

RECOGNISE THE BURDEN

- Review dry mouth, constipation, retention, confusion and falls, especially in older adults.
- Multiple medicines can produce a cumulative effect.



NURSE'S NOTE

A mnemonic can prompt assessment, but it cannot diagnose toxicity.

ACT / CHECK

Acute delirium, hyperthermia, marked tachycardia or retention with suspected toxicity requires urgent review.

Access for free at <https://openstax.org/books/anatomy-and-physiology-2e/pages/1-introduction>

Sources: B6 G6

Full references at the back

Medication administration

Connect every dose with its indication, checks and follow-up.



BEFORE

- Verify patient identifiers, prescription, medicine/formulation, dose, route and time.
- Check allergies and reaction details, indication, observations, labs and contraindications.

DURING

- Explain and obtain consent.
- Use approved preparation and administration references.
- Check calculations, compatibility, dilution, rate and device.
- Apply required independent checks.

AFTER

- Document after administration.
- Record omissions and reasons.
- Assess effect and adverse reactions.
- Follow up concerns; support the person's right to ask questions or decline.

HIGH-RISK / APINCHS

- A - Antimicrobials
- P - Potassium and other electrolytes
- I - Insulin
- N - Narcotics/opioids and other sedatives
- C - Chemotherapeutic agents
- H - Heparin and other anticoagulants
- S - Systems for safe administration



NURSE'S NOTE

Checking the "rights" helps, but does not replace clinical judgement or safe systems.

ACT / CHECK

If the order, dose or patient condition raises concern, resolve it before administration. Never give IV potassium as a direct push.

Medication calculations

Convert units first. Calculate, then check whether the answer is plausible.



Calculation	Rule and worked example
Metric conversions	$1 \text{ g} = 1,000 \text{ mg}$ $1 \text{ mg} = 1,000 \text{ micrograms}$ $1 \text{ L} = 1,000 \text{ mL}$
Tablet dose	$\text{Tablets} = \text{prescribed dose} \div \text{strength per tablet}$ Example: $500 \text{ mg} \div 250 \text{ mg} = 2 \text{ tablets}$
Liquid dose	$\text{mL} = \text{prescribed dose} \div \text{stock dose} \times \text{stock volume}$ Example: $300 \text{ mg} \div 150 \text{ mg} \times 5 \text{ mL} = 10 \text{ mL}$
Infusion rate	$\text{mL/hour} = \text{volume (mL)} \div \text{time (hours)}$ Example: $1,000 \text{ mL} \div 8 \text{ hours} = 125 \text{ mL/hour}$
Gravity infusion	$\text{Drops/min} = \text{volume (mL)} \times \text{set factor (drops/mL)} \div \text{time (min)}$ Example: $100 \times 20 \div 60 = 33 \text{ drops/min (rounded)}$

PAUSE & THINK What will you verify, and how will you assess the response?



NURSE'S NOTE

Use 0.5 mg, not .5 mg; use 5 mg, not 5.0 mg. Write "units" and "micrograms" clearly.

ACT / CHECK

Examples are arithmetic practice, not prescriptions. Measure medicines in mL; never assume a household spoon or a universal drops-to-mL conversion.

Acute coronary syndrome

A time-critical reduction in coronary blood flow may threaten myocardium.



RECOGNISE

- Chest pressure, tightness or discomfort.
- Pain in arm, jaw, back or epigastrium.
- Dyspnoea, sweating or nausea.
- Syncope or unexplained fatigue.
- Chest pain can be absent.

FIRST RESPONSE

1. Escalate immediately.
2. Assess ABCDE and vital signs.
3. Obtain a 12-lead ECG for prompt interpretation, ideally within 10 minutes of first clinical contact.
4. Follow the local ACS pathway.

DIAGNOSIS

- Interpret ECG, symptoms and serial high-sensitivity troponin together.
- One normal ECG or early troponin does not exclude ACS.
- Repeat assessment if symptoms persist or recur.

NURSING PRIORITIES

- Record symptom onset and evolution.
- Prepare monitoring and IV access as indicated.
- Give medicines under the approved pathway/prescription.
- Check bleeding risk and contraindications.
- Reperfusion decisions need urgent medical review.



NURSE'S NOTE

STEMI/NSTEMI labels do not tell you the full extent of threatened myocardium.

ACT / CHECK

Give oxygen for hypoxaemia or another clinical indication; do not automatically give high-flow oxygen for every chest-pain presentation.

Type 2 diabetes

Insulin resistance plus progressive loss of effective insulin secretion.



WHAT HAPPENS

- Glucose remains in the bloodstream when insulin action and supply are inadequate.
- Over time this can damage blood vessels, nerves, kidneys and eyes.

WHAT YOU MAY SEE

- Thirst, frequent urination, fatigue, blurred vision, infections or slow wound healing.
- Some people have no symptoms.
- Diagnosis uses appropriate blood tests and clinical assessment.

WARD PRIORITIES

- Check the glucose monitoring and treatment plan.
- Coordinate meals and medicines.
- Assess hydration, feet/skin, infection and renal function.
- Ask about usual medicines and self-management.

ACUTE ILLNESS

- Stress and corticosteroids may increase glucose.
- Vomiting, poor intake or fasting can change medicine needs.
- Obtain an individual sick-day/perioperative plan; do not make blanket changes.



NURSE'S NOTE

Use mmol/L for glucose in Australian practice. Inpatient targets are individualised.

ACT / CHECK

Vomiting, abdominal pain, dehydration, deep breathing or drowsiness needs urgent review. SGLT2-related ketoacidosis can occur without marked hyperglycaemia.

Hypoglycaemia

Blood glucose below 4 mmol/L requires prompt action.



RECOGNISE

- Sweating, shaking, hunger, palpitations, pallor, irritability or confusion.
- Severe episodes may cause seizures or loss of consciousness.
- Check glucose promptly.

AWAKE & SAFE TO SWALLOW

- Follow the local hypo pathway.
- Common adult approach: give about 15 g fast-acting carbohydrate.
- Recheck glucose after 15 minutes and repeat if still below 4 mmol/L.

UNSAFE SWALLOW / UNCONSCIOUS

- Call for urgent help.
- Do not give food or drink by mouth.
- Support airway and breathing; give IV glucose or glucagon only under the approved protocol and your scope.

AFTER RECOVERY

- Once glucose is above 4 mmol/L, provide the next meal or longer-acting carbohydrate as appropriate.
- Continue monitoring.
- Find the cause and review the medicine/meal plan.

PAUSE & THINK How might food intake or acute illness change your assessment?



NURSE'S NOTE

Use measured glucose products or the local protocol to deliver the intended carbohydrate amount.

ACT / CHECK

Recurrence can occur after insulin or sulfonylureas. A brief improvement does not end monitoring.

Diabetes medicines

Learn the class, the mechanism and the main nursing risk.



Class / example	Main action	Watch for
Metformin	Reduces hepatic glucose output	GI effects; review renal function and acute illness.
Sulfonylurea e.g. gliclazide	Stimulates insulin release	Hypoglycaemia, especially with poor intake.
SGLT2 inhibitor e.g. empagliflozin	Increases urinary glucose loss	Genital fungal infection, volume depletion; ketoacidosis risk.
GLP-1 receptor agonist e.g. semaglutide	Glucose-dependent insulin effect; slows gastric emptying	GI symptoms; check perioperative and fasting plan.
DPP-4 inhibitor e.g. sitagliptin	Enhances incretin action	Check renal dose requirements for the selected drug.
Insulin	Replaces/supplements insulin	Hypoglycaemia; match product, dose and timing.



NURSE'S NOTE

Combination tablets can contain more than one active ingredient. Check every ingredient.

ACT / CHECK

For a patient taking an SGLT2 inhibitor who is unwell, consider blood ketones even if glucose is not very high.

Insulin essentials

Know which part of the regimen covers meals and which covers background needs.



Type	Examples	Nursing implication
Rapid acting	Aspart, lispro, glulisine	Mealtime/correction insulin; exact timing is product-specific.
Short acting	Neutral / regular insulin	Usually needs a longer pre-meal interval; may be used IV under a specific protocol.
Intermediate	Isophane / NPH	Background action with a peak; cloudy suspension needing correct resuspension.
Long acting	Glargine and other basal products	Background insulin; duration and concentration vary.
Mixed / co-formulated	More than one action profile	Meal timing and mixing rules are product-specific.

CHECK EVERY TIME

- Verify product, concentration, dose in units, device, glucose, timing, food and recent doses.
- Rotate sites, avoid lipohypertrophy and follow device priming instructions.
- Never share pens.



NURSE'S NOTE

Clear appearance does not identify an insulin. Do not mix products unless specifically permitted.

ACT / CHECK

People with type 1 diabetes require basal insulin even when fasting; obtain an adjusted plan rather than simply omitting it.

Metformin

A common medicine for type 2 diabetes; usually low hypo risk when used alone.



MAIN ACTION

- Reduces glucose production by the liver and improves insulin sensitivity.
- It is often an early treatment option, but the overall regimen depends on comorbidities and goals.

COMMON EFFECTS

- Nausea, diarrhoea and abdominal discomfort.
- Taking it with food may improve tolerance.
- Check immediate-release versus modified-release formulation; do not crush modified-release tablets.

MONITOR

- Renal function, glucose and GI tolerance.
- Long-term use may lower vitamin B12; consider this with anaemia or neuropathy and follow the review plan.

ACUTE ILLNESS

- Dehydration, acute kidney injury, hypoxia or severe illness requires prompt review of suitability.
- Contrast/procedure instructions depend on renal function and local guidance.



NURSE'S NOTE

Metformin alone rarely causes hypoglycaemia; combination treatment can change the risk.

ACT / CHECK

Lactic acidosis is rare but serious. Marked weakness, breathlessness, persistent vomiting or severe illness warrants urgent assessment.

Respiratory medicines

Reliever and preventer roles can overlap in combination inhalers.



Medicine / class	Main role	Nursing point
Salbutamol	Short-acting beta-2 agonist	Rapid bronchodilation; may cause tremor, tachycardia and hypokalaemia at high doses.
Inhaled corticosteroid e.g. budesonide, fluticasone	Reduces airway inflammation	Monitor technique, adherence, oral thrush and dysphonia.
ICS-formoterol	Anti-inflammatory plus rapid bronchodilation	Some prescribed products serve as both reliever and maintenance therapy.
Antimuscarinic e.g. ipratropium, tiotropium	Blocks muscarinic airway effects	Dry mouth; device/eye exposure and urinary retention precautions.

PAUSE & THINK How will you assess breathing and the response to treatment?



NURSE'S NOTE

In adult/adolescent asthma, an ICS-containing plan is central. Do not infer purpose from inhaler colour.

ACT / CHECK

Increasing reliever use, inability to speak comfortably, exhaustion or reduced air entry requires prompt escalation.

Inhalers: sequence & technique

Check the written regimen and the exact device.



WHEN TWO ARE PRESCRIBED

- If a separate short-acting bronchodilator is indicated at the same time as a maintenance corticosteroid, it is commonly given first.
- There is no universal sequence or waiting interval for all inhalers.

MATCH THE DEVICE

- A pressurised metered-dose inhaler usually requires a slow inhalation and coordination; a spacer may help.
- A dry-powder inhaler generally needs a stronger inhalation.
- Demonstrate the manufacturer's steps.

AFTER MAINTENANCE ICS

- Rinse the mouth and spit out after regular corticosteroid doses to reduce local adverse effects.
- A compatible spacer also helps with pressurised inhalers.

CHECK UNDERSTANDING

- Ask the person to show their technique.
- Check dose counter, priming, cleaning and storage.
- Explain which inhaler is used for which purpose and the action plan for worsening symptoms.



NURSE'S NOTE

Rinsing is not routinely required after as-needed low-dose budesonide-formoterol alone; follow the prescribed plan.

ACT / CHECK

Never delay needed reliever treatment while arranging mouth rinsing or another inhaler.

Systemic corticosteroids

Reduce inflammation and immune activity; risks rise with dose and duration.



EXAMPLES & EFFECT

- Prednisolone, dexamethasone and hydrocortisone.
- Used for many inflammatory conditions and, in some settings, hormone replacement.

SHORT-TERM WATCH POINTS

- Raised glucose, mood or sleep change, dyspepsia, fluid retention and blood-pressure changes.
- Infection can develop or be less obvious.

LONGER-TERM RISKS

- Adrenal suppression, osteoporosis, muscle weakness, thin skin and increased infection risk.
- Review preventive measures and monitoring in the treatment plan.

ADMINISTRATION & REVIEW

- Check dose, duration and any taper.
- Oral doses are often taken with food; timing follows the regimen.
- Monitor glucose and fluid status when relevant.
- Ask about recent steroid courses.



NURSE'S NOTE

Steroid-related glucose rises may be greatest later in the day; follow the prescribed monitoring schedule.

ACT / CHECK

Do not abruptly stop prolonged systemic therapy. Tapering and stress-dose needs require a prescriber's plan; not every short course needs a taper.

Paracetamol

Analgesic and antipyretic.



MAIN USE

- Relieves pain and fever.
- It has little anti-inflammatory effect.
- Oral, rectal and IV formulations all contribute to the total exposure.

BEFORE EACH DOSE

- Check the prescription, last dose and total over the previous 24 hours.
- Look for paracetamol in combination analgesics, cold/flu products and medicines brought from home.

DOSE AWARENESS

- For many adults, the usual maximum is 4 g in 24 hours, but a lower prescribed limit may be needed with low body weight, frailty, malnutrition or liver risk.
- Check formulation-specific intervals.

SUSPECTED OVERDOSE

- Obtain urgent medical/toxicology advice even if the person feels well.
- Acetylcysteine is the antidote.
- Treatment depends on timing, formulation, blood tests and the overdose pattern.

PAUSE & THINK How will you assess benefit and recognise adverse effects?



NURSE'S NOTE

The label strength is not the amount already given today: add all routes and all products.

ACT / CHECK

Overdose can cause severe liver injury with few early symptoms. Do not wait for pain or jaundice.

NSAIDs

Non-steroidal anti-inflammatory medicines: pain relief with important risks.



EXAMPLES

- Ibuprofen, naproxen, diclofenac; ketorolac is used for short-term treatment under product-specific limits.
- Check whether more than one NSAID is prescribed or self-administered.

MAJOR RISKS

- GI ulceration or bleeding, acute kidney injury, fluid retention and cardiovascular events.
- Some people develop bronchospasm or other hypersensitivity reactions.

BEFORE ADMINISTRATION

- Review renal function, hydration, ulcer/bleeding history, heart failure, blood pressure, pregnancy and concurrent medicines.
- Anticoagulants or antiplatelets increase bleeding concerns.

MONITOR & TEACH

- Assess pain response, abdominal symptoms, fluid status, BP and renal function as indicated.
- Food may reduce dyspepsia but does not prevent serious GI bleeding.

PAUSE & THINK How will you assess benefit and recognise adverse effects?



NURSE'S NOTE

NSAID + ACE inhibitor/ARB + diuretic can increase acute kidney injury risk, especially during dehydration.

ACT / CHECK

Black stools, haematemesis, reduced urine, worsening dyspnoea or swelling needs prompt review.

Opioid safety

Pain relief must be balanced with ventilation, alertness and function.



EXAMPLES

- Morphine, oxycodone and fentanyl.
- Formulations and routes differ greatly in onset and duration; they are not dose-equivalent.

ASSESS BEFORE & AFTER

- Pain and functional goal; sedation/consciousness; respiratory rate, depth and effort; SpO₂ and BP.
- Review renal function, opioid tolerance and other sedating medicines.

COMMON PROBLEMS

- Nausea, constipation, dizziness, urinary retention and sedation.
- Plan bowel care and falls precautions as indicated.
- Alcohol, benzodiazepines and other sedatives can compound harm.

SUSPECTED TOXICITY

- Withhold further opioid, call for urgent help and support airway/ventilation.
- Give naloxone under the approved protocol.
- Continue monitoring because opioid effects can outlast naloxone.



NURSE'S NOTE

Supplemental oxygen may preserve SpO₂ while ventilation worsens. Sedation is an important early warning.

ACT / CHECK

Difficult to rouse, slow/shallow breathing or apnoea is an emergency. Fentanyl patches are not for acute pain in opioid-naïve patients.

Cardiovascular medicines

Link the therapeutic effect to the bedside assessment.



Class	Main effect/examples	Nursing focus
ACE inhibitor / ARB	Lowers BP; selected cardiac/renal benefits	BP, potassium, renal function
Beta blocker	Slows rate; reduces cardiac workload	Pulse, BP, conduction and symptoms
Calcium channel blocker	Amlodipine: vasodilation	BP and peripheral oedema
Rate-limiting CCB	Verapamil / diltiazem: slows conduction	Pulse, BP and interacting rate-slowing drugs
Diuretic	Promotes sodium/water loss	Fluid balance, weight, electrolytes, renal function
Nitrate	Glyceryl trinitrate (GTN): vasodilation	BP, headache, dizziness and contraindications
Statin	Atorvastatin: lowers LDL and vascular risk	Muscle symptoms and liver tests when indicated



NURSE'S NOTE

Not all calcium channel blockers slow heart rate, and not all cardiovascular drugs share the same hold criteria.

ACT / CHECK

GTN with recent PDE-5 inhibitor use can cause severe hypotension. Verify the medicine and required interval.

ACE inhibitors vs ARBs

Both act on the renin-angiotensin system, at different points.



Feature	ACE inhibitor	ARB
Name pattern	-pril	-sartan
Examples	Ramipril, perindopril, enalapril	Candesartan, irbesartan, losartan
Action	Reduces angiotensin II formation	Blocks angiotensin II receptor effects
Common concerns	Hypotension, raised potassium, renal function change	Hypotension, raised potassium, renal function change
Distinguishing issue	Dry cough; angioedema can occur	Cough is less common; angioedema is still possible

MONITOR

- BP, symptoms, potassium and renal function, especially after initiation/dose changes or acute illness.
- Review potassium supplements, salt substitutes, NSAIDs and other interacting medicines.



NURSE'S NOTE

Do not routinely combine an ACE inhibitor with an ARB. Sacubitril/valsartan has separate transition requirements.

ACT / CHECK

Lip, tongue or throat swelling needs emergency assessment. Both classes are avoided in pregnancy.

Beta blockers

Less adrenergic stimulation usually means a slower, less forceful heartbeat.



EXAMPLES

- Metoprolol, bisoprolol, atenolol and propranolol.
- Many names end in -olol, but there are exceptions such as carvedilol.

USE & EFFECT

- Selected agents are used for heart failure, angina, arrhythmias and after myocardial infarction.
- Effects include lower heart rate and blood pressure and reduced cardiac workload.

BEFORE & AFTER

- Assess pulse/rhythm, BP, dizziness, perfusion and heart-failure symptoms.
- Use the patient's prescribed parameters; seek review for bradycardia, hypotension or deterioration.

WATCH FOR

- Fatigue, bradycardia, hypotension and bronchospasm.
- Non-selective agents carry greater airway concern.
- Beta blockers may mask some adrenergic signs of hypoglycaemia.



NURSE'S NOTE

A cardioselective beta blocker is not completely free of bronchospasm risk.

ACT / CHECK

Do not abruptly stop regular therapy without a plan. Symptomatic bradycardia, syncope or new wheeze requires prompt review.

Furosemide & diuresis

A loop diuretic that increases renal sodium and water excretion.



WHY IT IS GIVEN

- Often used for fluid overload, including oedema associated with heart failure.
- You may also see the older Australian spelling "frusemide".

MONITOR RESPONSE

- Breathing, oedema, urine output, fluid balance and weight under consistent conditions.
- Check BP, symptoms, renal function, sodium, potassium and magnesium as ordered.

ADVERSE EFFECTS

- Volume depletion, postural hypotension, electrolyte loss and renal function deterioration.
- Hearing toxicity is a concern with high exposure or rapid IV administration.

PRACTICAL CARE

- Check IV rate/dilution in the injectable medicines guide.
- Support safe toileting and falls prevention.
- Oral timing should suit the plan and access to a toilet.
- Do not automatically add potassium.



NURSE'S NOTE

More urine is not the whole outcome: assess whether congestion improves without excessive depletion.

ACT / CHECK

New dizziness, hypotension, oliguria, major electrolyte disturbance or hearing symptoms needs review.

Anticoagulant monitoring

Preventing clots does not remove the need to assess bleeding.



Group	Monitoring	Reversal principle
Unfractionated heparin	Therapeutic infusions: aPTT or anti-Xa per protocol; platelets/bleeding.	Protamine reverses heparin.
Low molecular weight heparin	e.g. enoxaparin: renal function, weight, platelets; anti-Xa in selected cases.	Protamine only partly reverses effect.
Warfarin	INR target is indication-specific; check interactions and prescribed daily dose.	Vitamin K; major bleeding may need prothrombin complex concentrate.
DOACs	Apixaban, rivaroxaban, dabigatran: renal function, haemoglobin, bleeding, adherence.	Specific reversal/support depends on drug, severity and local access.



NURSE'S NOTE

Routine INR does not measure DOAC activity. "No routine INR" does not mean "no monitoring".

ACT / CHECK

Major bleeding, new severe headache, neurological change or a significant head injury needs urgent assessment. Reversal is specialist-directed.

Antibiotic safety

Right indication, timely treatment and active monitoring.



BEFORE THE DOSE

- Check allergies and reaction details, cultures when ordered, renal function, dose, route and compatibility.
- Obtain specimens promptly; do not delay time-critical treatment to collect them.

AMINOGLYCOSIDES

- Gentamicin and related agents can cause renal, hearing and vestibular toxicity.
- Monitor renal function and symptoms such as tinnitus, hearing change or imbalance.
- Drug-level timing must match the regimen.

VANCOMYCIN

- Monitor renal function and prescribed drug levels.
- AUC-based monitoring is preferred for many serious infections.
- Rapid IV administration can cause flushing, itching and hypotension; follow the ordered infusion rate.

FOLLOW THROUGH

- Record exact dose, infusion and sample times.
- Review clinical response and adverse effects.
- Ongoing aminoglycoside or vancomycin treatment commonly needs therapeutic drug monitoring beyond 48 hours.



NURSE'S NOTE

A drug level without reliable dose/sample timing can be misleading.

ACT / CHECK

For suspected infusion reaction: stop the infusion and assess urgently. Airway, breathing or circulation compromise requires emergency response.

Antiarrhythmic essentials

These medicines can also cause arrhythmias: monitoring is essential.



Medicine	High-priority nursing concern
Amiodarone	Bradycardia/conduction effects; pulmonary, thyroid and liver toxicity. Check ECG, relevant labs and new respiratory symptoms.
Adenosine	Very short acting; used for selected rhythms under a monitored protocol. Transient flushing, chest discomfort or AV block can occur.
Lidocaine (lignocaine)	Selected ventricular arrhythmias. Watch for CNS toxicity: confusion, twitching or seizures; also hypotension/conduction effects.
Sotalol	Beta-blocking and antiarrhythmic actions. QT prolongation/torsades risk; check ECG, renal function, potassium and magnesium.
Procainamide	Specialist use. Hypotension and QRS/QT effects; longer exposure can cause a lupus-like syndrome.

PAUSE & THINK Which observations connect this concept to your patient?



NURSE'S NOTE

Check rhythm, haemodynamic stability, interactions and prescribed monitoring before administration.

ACT / CHECK

Never apply one antiarrhythmic protocol to every tachycardia. Unstable rhythms require urgent escalation.

Digoxin

Increases contractility and slows AV nodal conduction.



BEFORE ADMINISTRATION

- Assess heart rate and rhythm using the approved method.
- Review prescribed parameters, renal function, potassium, magnesium and interacting medicines.

TOXICITY CLUES

- Anorexia, nausea/vomiting, confusion, visual disturbance, bradycardia or other arrhythmias.
- Toxicity may occur even when a measured level falls in the laboratory therapeutic range.

WHY THE RISK CHANGES

- Renal impairment, dehydration and interacting drugs can increase exposure.
- Low potassium or magnesium increases susceptibility; severe acute toxicity may instead produce hyperkalaemia.

LEVELS & RESPONSE

- If a level is ordered, document the last dose and sample time; samples are generally taken at least 6 hours after a dose.
- Suspected toxicity requires urgent review, ECG and electrolyte assessment.

PAUSE & THINK Which observations connect this concept to your patient?



NURSE'S NOTE

An apical pulse is assessed over the cardiac apex with a stethoscope; palpating the neck measures a carotid pulse.

ACT / CHECK

Withhold and urgently seek review if toxicity is suspected. Digoxin-specific antibody fragments (Fab) may be required in severe toxicity.

Thyroid medicines

Replacement for low hormone levels; suppression for excess production.



LEVOTHYROXINE

- Replaces thyroxine (T4) in hypothyroidism.
- Take consistently according to the product instructions, commonly before breakfast on an empty stomach.

REPLACEMENT MONITORING

- Follow TSH/free T4 monitoring.
- Excess replacement may cause palpitations, tremor, insomnia or heat intolerance.
- Calcium and iron can reduce absorption; check the required separation interval.

ANTITHYROID MEDICINES

- Carbimazole and propylthiouracil reduce thyroid hormone production.
- Monitor thyroid function and adverse effects; pregnancy and treatment selection require specialist guidance.

SAFETY EDUCATION

- Fever, sore throat or mouth ulcers may indicate agranulocytosis.
- Stop carbimazole and seek urgent medical advice/blood count under the treatment instructions.
- Jaundice or dark urine also requires urgent review.



NURSE'S NOTE

Australian references commonly use carbimazole; US charts often list methimazole.

ACT / CHECK

Severe palpitations, collapse, marked fever or altered consciousness in a person with thyroid disease needs emergency assessment.

Medicine name patterns

Useful clues to prompt a lookup, never proof of identity or indication.



Pattern	Class clue	Example
-pril	ACE inhibitor	Ramipril
-sartan	ARB	Candesartan
-olol	Many beta blockers	Metoprolol
-dipine	Dihydropyridine CCB	Amlodipine
-statin	Lipid-lowering statin*	Atorvastatin
-floxacin	Fluoroquinolone	Ciprofloxacin
-cycline	Tetracycline group	Doxycycline
-prazole	Proton pump inhibitor*	Pantoprazole
-tidine	H2-receptor blocker	Famotidine
-setron	5-HT ₃ antagonist	Ondansetron
-azepam / -azolam	Many benzodiazepines	Diazepam / midazolam
-terol	Many beta-2 agonists	Formoterol
-lukast	Leukotriene antagonist	Montelukast
-gliflozin	SGLT2 inhibitor	Empagliflozin



NURSE'S NOTE

*Exceptions matter: nystatin is an antifungal; aripiprazole is an antipsychotic. Salbutamol does not end in -terol.

ACT / CHECK

"-mycin" spans different antibiotic classes and is not a reliable mechanism or safety guide.

Abbreviations & credits

Decode the language and trace the anatomy illustrations.



Abbreviation	Meaning
ABG / VBG	Arterial / venous blood gas
ACS / MI	Acute coronary syndrome / myocardial infarction
BP / HR / RR	Blood pressure / heart rate / respiratory rate
BGL / DKA	Blood glucose level / diabetic ketoacidosis
DOAC / INR	Direct-acting oral anticoagulant / international normalised ratio
aPTT / AUC	Activated partial thromboplastin time / area under the concentration-time curve
ECG / GCS	Electrocardiogram / Glasgow Coma Scale
eGFR / GI	Estimated glomerular filtration rate / gastrointestinal
FiO ₂ / SpO ₂	Inspired oxygen fraction / oxygen saturation measured by pulse oximetry
ICS / AV	Inhaled corticosteroid / atrioventricular
IV / PIVC	Intravenous / peripheral intravenous catheter
TSH / T ₄	Thyroid-stimulating hormone / thyroxine

ANATOMY & REUSE

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Reference directory

Source codes link each topic to supporting guidance. | 1 of 7



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E9

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Healthdirect: type 2 diabeteswww.healthdirect.gov.au/type-2-diabetes

F2

Diabetes Australia: hypoglycaemia and hyperglycaemiawww.diabetesaustralia.com.au/managing-diabetes/hypo-hyperglycaemia/

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Healthdirect: diabetes medicineswww.healthdirect.gov.au/diabetes-medicines

F4

NDSS: all about insulinwww.ndss.com.au/managing-diabetes/insulin-and-medicines/all-about-insulin/

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Healthdirect: metforminwww.healthdirect.gov.au/metformin

F6

Australian Diabetes Society: SGLT2 inhibitor DKA alertwww.diabetessociety.com.au/downloads/202202092021ADS_DKA_SGLT2i_Alert_highlightedchanges_Jan22.pdf

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Healthdirect: thyroid medicineswww.healthdirect.gov.au/thyroid-medicines

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ACSQHC Medicine Finder: Neo-Mercazole (carbimazole)www.safetyandquality.gov.au/medicine-finder/neo-mercazole-tablets

G1

Australian Asthma Handbook: medicines guidewww.astmahandbook.org.au/medicines-guide

Sources consulted September 2026. Some foundational references are older; check current product information and local policy at the point of care. External links may change.

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ANZCA: position statement on acute pain management (2023)

[www.anzca.edu.au/getContentAsset/136f5a83-d1d0-4f34-be72-87b62b721d14/80feb437-d24d-46b8-a858-4a2a28b9b970/PS41\(G\)-Acute-pain-2023.pdf?language=en](http://www.anzca.edu.au/getContentAsset/136f5a83-d1d0-4f34-be72-87b62b721d14/80feb437-d24d-46b8-a858-4a2a28b9b970/PS41(G)-Acute-pain-2023.pdf?language=en)

Sources consulted September 2026. Some foundational references are older; check current product information and local policy at the point of care. External links may change.

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**C9****Australian Prescriber: acute oxygen use and delivery devices**australianprescriber.tg.org.au/articles/acute-use-of-oxygen-therapy.html**B8****OpenStax: anatomical terminology; Figure 1.13**openstax.org/books/anatomy-and-physiology-2e/pages/1-6-anatomical-terminology**A9****Jonathan Moore: arterial ulcer photograph; CC BY 3.0**commons.wikimedia.org/wiki/File:Arterial_ulcer_peripheral_vascular_disease.jpg**A10****Nini00: venous ulcer photograph; CC BY-SA 3.0**commons.wikimedia.org/wiki/File:%C3%A9Alceras_antes_da_cirurgia.JPG

Sources consulted September 2026. Some foundational references are older; check current product information and local policy at the point of care. External links may change.